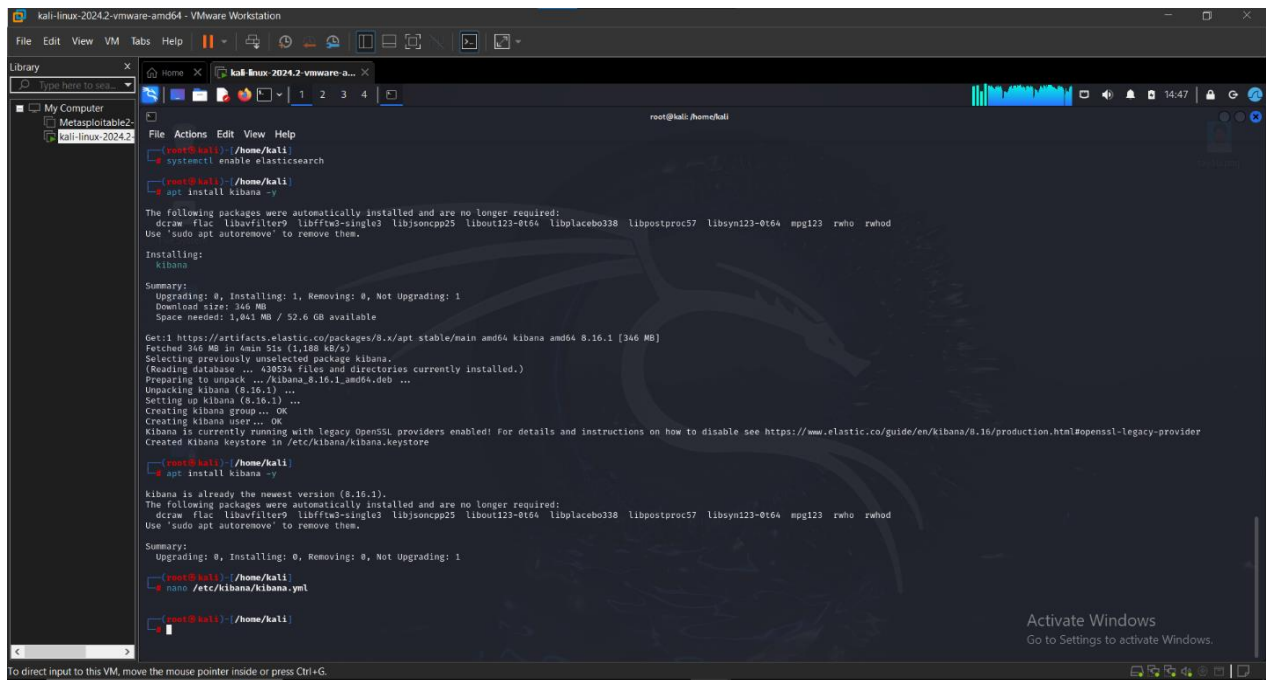


Name: Muhammad Tayyab Nisar

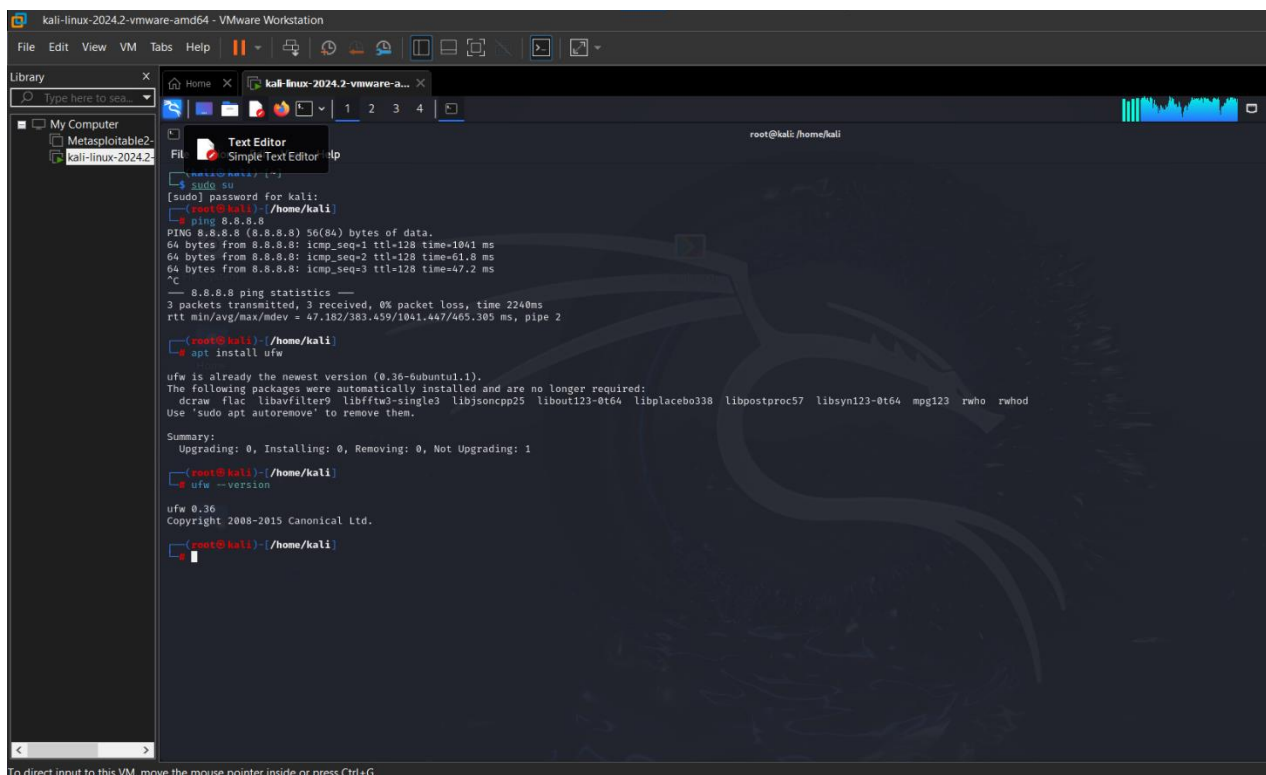
Task 1

```
kali-linux-2024.2-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
Library
My Computer
Metasploitable2
kali-linux-2024.2
root@kali: /home/kali
Minimize all open windows and show the desktop
root@kali: /home/kali
# apt install elasticsearch -y
The following packages were automatically installed and are no longer required:
dcraw flac libavfilter9 libfftw3-single3 libjsoncpp25 libout123-0t64 libplacebo338 libpostproc57 libsyn123-0t64 mpg123 rwho rwho
Use 'sudo apt autoremove' to remove them.
Installing:
elasticsearch
Summary:
Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 1
Download size: 635 MB
Space needed: 1,269 MB / 54.5 GB available
Get:1 https://artifacts.elastic.co/packages/6.x/apt stable/main amd64 elasticsearch amd64 8.10.1 [635 MB]
Fetched 572 MB in 7min 38s (1,247 kB/s)
Selecting previously unselected package elasticsearch.
(Reading database ... 414589 files and directories currently installed.)
Preparing to unpack .../elasticsearch_8.10.1_amd64.deb ...
Creating elasticsearch group ... OK
Creating elasticsearch user ... OK
Unpacking elasticsearch (8.10.1) ...
Setting up elasticsearch (8.10.1) ...
----- Security autoconfiguration information -----
Authentication and authorization are enabled.
TLS for the transport and HTTP layers is enabled and configured.
The generated password for the elastic built-in superuser is : gwcnb756j+De6LjY-4c0
If this node should join an existing cluster, you can reconfigure this with
'/usr/share/elasticsearch/bin/elasticsearch-reconfigure-node --enrollment-token <token-here>'
after creating an enrollment token on your existing cluster.
You can complete the following actions at any time:
Reset the password of the elastic built-in superuser with
'/usr/share/elasticsearch/bin/elasticsearch-reset-password -u elastic'.
Generate an enrollment token for Kibana instances with
'/usr/share/elasticsearch/bin/elasticsearch-create-enrollment-token -s kibana'.
Generate an enrollment token for Elasticsearch nodes with
'/usr/share/elasticsearch/bin/elasticsearch-create-enrollment-token -s node'.
## NOT starting on installation, please execute the following statements to configure elasticsearch service to start automatically using systemd
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.
```

```
kali-linux-2024.2-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
Library
My Computer
Metasploitable2
kali-linux-2024.2
root@kali: /home/kali
File Actions Edit View Help
root@kali: /home/kali
# apt install elasticsearch -y
elasticsearch is already the newest version (8.10.1).
The following packages were automatically installed and are no longer required:
dcraw flac libavfilter9 libfftw3-single3 libjsoncpp25 libout123-0t64 libplacebo338 libpostproc57 libsyn123-0t64 mpg123 rwho rwho
Use 'sudo apt autoremove' to remove them.
Summary:
Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 1
root@kali: /home/kali
# nano /etc/elasticsearch/elasticsearch.yml
root@kali: /home/kali
# nano /etc/elasticsearch/elasticsearch.yml
root@kali: /home/kali
# systemctl restart elasticsearch
root@kali: /home/kali
# systemctl status elasticsearch
● elasticsearch.service - Elasticsearch
Loaded: loaded (/usr/lib/systemd/system/elasticsearch.service; disabled; preset: disabled)
Active: active (running) since Fri 2024-12-06 11:29:28 EST; 56s ago
Invocation: 66412b027547c29faccdeab4c1e1
Docs: https://www.elastic.co
Main PID: 33050 (java)
Tasks: 185 (limit: 3388)
Memory: 1.7G (peak: 1.7G swap: 256M swap peak: 256.2M)
CPU: 47.81s
CGroup: /system.slice/elasticsearch.service
┌─33050 /usr/share/elasticsearch/jdk/bin/java -Xms4m -Xmx64m -XX:UseSerialGC -Dcli.name=server -Dcli.script=/usr/share/elasticsearch/bin/elasticsearch -Dcli.lib=/usr/share/elasticsearch/bin/elasticsearch -Dcli.lib=/usr/share/elasticsearch/bin/elasticsearch -Dcli.path.home=/usr/share/elasticsearch/jdk/bin/java -Des.networkaddress.cache.ttl=60 -Des.networkaddress.cache.negative.ttl=10 -Djava.security.manager=allow -XX:+AlwaysPreTouch -Xss1m -Djava.net.preferIPv4Stack=true
└─33192 /usr/share/elasticsearch/modules/*-pack-nl/platform/linux-x86_64/bin/controller
Dec 06 11:29:59 kali systemd[1]: Starting elasticsearch.service - Elasticsearch ...
Dec 06 11:29:04 kali systemd-entypoint[33125]: CompileCommand: dontinline java/lang/invoke/MethodHandle.setAsTypeCache bool dontinline = true
Dec 06 11:29:04 kali systemd-entypoint[33125]: CompileCommand: dontinline java/lang/invoke/MethodHandle.setAsTypeCache bool dontinline = true
Dec 06 11:29:28 kali systemd[1]: Started elasticsearch.service - Elasticsearch.
lines 1-36/36 (END)
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.
```

Task 2



```
(root@kali)-[/home/kali]
# ping 150.1.7.104

PING 150.1.7.104 (150.1.7.104) 56(84) bytes of data.
64 bytes from 150.1.7.104: icmp_seq=1 ttl=64 time=1.15 ms
64 bytes from 150.1.7.104: icmp_seq=2 ttl=64 time=0.456 ms
64 bytes from 150.1.7.104: icmp_seq=3 ttl=64 time=0.384 ms
64 bytes from 150.1.7.104: icmp_seq=4 ttl=64 time=0.386 ms
^C
 150.1.7.104 ping statistics —
4 packets transmitted, 4 received, 0% packet loss, time 3396ms
rtt min/avg/max/mdev = 0.384/0.594/1.151/0.322 ms

(root@kali)-[/home/kali]
# ufw allow from 150.1.7.104 to any port 22

Rule added

(root@kali)-[/home/kali]
# ufw deny 22

Rule added
Rule added (v6)

(root@kali)-[/home/kali]
# ufw allow 80
ufw allow 443

Rule added
Rule added (v6)
Skipping adding existing rule
Skipping adding existing rule (v6)

(root@kali)-[/home/kali]
#
```

Use the mouse pointer inside or press Ctrl+G

```
(root@kali)-[/home/kali]
# ufw enable

Firewall is active and enabled on system startup

(root@kali)-[/home/kali]
# sudo ufw default deny incoming
sudo ufw default allow outgoing

Default incoming policy changed to 'deny'
(be sure to update your rules accordingly)
Default outgoing policy changed to 'allow'
(be sure to update your rules accordingly)

(root@kali)-[/home/kali]
#
```

```
(root@kali)-[/home/kali]
# ufw disable
```

Firewall stopped and disabled on system startup

```
(root@kali)-[/home/kali]
#
```

```
(root@kali)-[/home/kali]
# nmap -p 22,80,443 150.1.7.104
```

Starting Nmap 7.94SVN (<https://nmap.org>) at 2024-12-08 14:35 EST
Nmap scan report for 150.1.7.104
Host is up (0.00041s latency).

```
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
443/tcp   closed https
MAC Address: 00:0C:29:23:96:F4 (VMware)
```

Nmap done: 1 IP address (1 host up) scanned in 0.38 seconds

```
(root@kali)-[/home/kali]
# tail -f /var/log/ufw.log
```

```
Dec 8 14:23:26 kali kernel: [ 253.652003] [UFW BLOCK] IN=eth1 OUT= MAC= SRC=150.1.7.101 DST=239.255.255.250 LEN=635 TOS=0x00 PREC=0x00 TTL=1 ID=37839 DF PROTO=UDP SPT=50821 DPT=3702 LEN=615
Dec 8 14:23:26 kali kernel: [ 253.654214] [UFW BLOCK] IN=eth1 OUT= MAC= SRC=fe80:0000:0000:0000:020c:29ff:fe37:911d DST=ff02:0000:0000:0000:0000:0000:0000:000c LEN=655 TC=0 HOPLIMIT=1 FLOWLBL=242307 PROTO=UDP SPT=51824 DPT=3702 LEN=615
Dec 8 14:23:26 kali kernel: [ 253.831625] [UFW BLOCK] IN=eth0 OUT= MAC= SRC=fe80:0000:0000:0000:020c:29ff:fe37:9113 DST=ff02:0000:0000:0000:0000:0000:0000:000c LEN=655 TC=0 HOPLIMIT=1 FLOWLBL=172575 PROTO=UDP SPT=53733 DPT=3702 LEN=615
Dec 8 14:23:26 kali kernel: [ 253.871174] [UFW BLOCK] IN=eth1 OUT= MAC= SRC=fe80:0000:0000:0000:020c:29ff:fe37:911d DST=ff02:0000:0000:0000:0000:0000:0000:000c LEN=655 TC=0 HOPLIMIT=1 FLOWLBL=242307 PROTO=UDP SPT=51824 DPT=3702 LEN=615
Dec 8 14:23:26 kali kernel: [ 253.873026] [UFW BLOCK] IN=eth0 OUT= MAC= SRC=192.168.23.128 DST=239.255.255.250 LEN=635 TOS=0x00 PREC=0x00 TTL=1 ID=8906 DF PROTO=UDP SPT=33253 DPT=3702 LEN=615
Dec 8 14:23:26 kali kernel: [ 253.895916] [UFW BLOCK] IN=eth1 OUT= MAC= SRC=150.1.7.101 DST=239.255.255.250 LEN=635 TOS=0x00 PREC=0x00 TTL=1 ID=37891 DF PROTO=UDP SPT=50821 DPT=3702 LEN=615
Dec 8 14:23:26 kali kernel: [ 254.004568] [UFW BLOCK] IN=eth0 OUT= MAC= SRC=192.168.23.128 DST=239.255.255.250 LEN=635 TOS=0x00 PREC=0x00 TTL=1 ID=8929 DF PROTO=UDP SPT=33253 DPT=3702 LEN=615
Dec 8 14:23:26 kali kernel: [ 254.006030] [UFW BLOCK] IN=eth0 OUT= MAC= SRC=fe80:0000:0000:0000:020c:29ff:fe37:9113 DST=ff02:0000:0000:0000:0000:0000:0000:000c LEN=655 TC=0 HOPLIMIT=1 FLOWLBL=172575 PROTO=UDP SPT=53733 DPT=3702 LEN=615
Dec 8 14:23:27 kali kernel: [ 254.086609] [UFW BLOCK] IN=eth1 OUT= MAC= SRC=fe80:0000:0000:0000:020c:29ff:fe37:911d DST=ff02:0000:0000:0000:0000:0000:0000:000c LEN=655 TC=0 HOPLIMIT=1 FLOWLBL=242307 PROTO=UDP SPT=51824 DPT=3702 LEN=615
Dec 8 14:23:27 kali kernel: [ 254.094055] [UFW BLOCK] IN=eth1 OUT= MAC= SRC=150.1.7.101 DST=239.255.255.250 LEN=635 TOS=0x00 PREC=0x00 TTL=1 ID=37896 DF PROTO=UDP SPT=50821 DPT=3702 LEN=615
```

Move the mouse pointer inside or press Ctrl+G.




```
(root@kali) ~/home/kali
```

```
# nmap -p- 150.1.7.101
```

```
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-12-08 14:34 EST
```

```
Nmap scan report for 150.1.7.101
```

```
Host is up (0.0000020s latency).
```

```
Not shown: 65534 closed tcp ports (reset)
```

```
PORT      STATE SERVICE
```

```
9200/tcp  open  wap-wsp
```

```
Nmap done: 1 IP address (1 host up) scanned in 11.67 seconds
```

```
(root@kali) ~/home/kali
```

```
#
```

move the mouse pointer inside or press Ctrl+G

```
(root@kali) ~/home/kali
```

```
# nmap -p- 150.1.7.104
```

```
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-12-08 14:33 EST
```

```
Nmap scan report for 150.1.7.104
```

```
Host is up (0.0019s latency).
```

```
Not shown: 65505 closed tcp ports (reset)
```

```
PORT      STATE SERVICE
```

```
21/tcp    open  ftp
```

```
22/tcp    open  ssh
```

```
23/tcp    open  telnet
```

```
25/tcp    open  smtp
```

```
53/tcp    open  domain
```

```
80/tcp    open  http
```

```
111/tcp   open  rpcbind
```

```
139/tcp   open  netbios-ssn
```

```
445/tcp   open  microsoft-ds
```

```
512/tcp   open  exec
```

```
513/tcp   open  login
```

```
514/tcp   open  shell
```

```
1099/tcp  open  rmiregistry
```

```
1524/tcp  open  ingreslock
```

```
2049/tcp  open  nfs
```

```
2121/tcp  open  ccproxy-ftp
```

```
3306/tcp  open  mysql
```

```
3632/tcp  open  distccd
```

```
5432/tcp  open  postgresql
```

```
5900/tcp  open  vnc
```

```
6000/tcp  open  X11
```

```
6667/tcp  open  irc
```

```
6697/tcp  open  ircs-u
```

```
8009/tcp  open  ajp13
```

```
8180/tcp  open  unknown
```

```
8787/tcp  open  msgsrvr
```

```
38853/tcp open  unknown
```

```
47829/tcp open  unknown
```

```
55848/tcp open  unknown
```

```
56347/tcp open  unknown
```

```
MAC Address: 00:0C:29:23:96:F4 (VMware)
```

```
Nmap done: 1 IP address (1 host up) scanned in 6.53 seconds
```

```
(root@kali) ~/home/kali
```

move the mouse pointer inside or press Ctrl+G

here to search



8°C Haze 10:35 PM 12/8/2024

```
(root@kali)-[/home/kali]
# ufw status numbered

Status: active

    To Action From
    --
[ 1] 22/tcp ALLOW IN Anywhere
[ 2] 2222 ALLOW IN Anywhere
[ 3] 80/tcp ALLOW IN Anywhere
[ 4] 443 ALLOW IN Anywhere
[ 5] 9200 ALLOW IN Anywhere
[ 6] 5601/tcp ALLOW IN Anywhere
[ 7] 22 ALLOW IN 150.1.7.104
[ 8] 22 DENY IN Anywhere
[ 9] 80 ALLOW IN Anywhere
[10] 22/tcp (v6) ALLOW IN Anywhere (v6)
[11] 2222 (v6) ALLOW IN Anywhere (v6)
[12] 80/tcp (v6) ALLOW IN Anywhere (v6)
[13] 443 (v6) ALLOW IN Anywhere (v6)
[14] 9200 (v6) ALLOW IN Anywhere (v6)
[15] 5601/tcp (v6) ALLOW IN Anywhere (v6)
[16] 22 (v6) DENY IN Anywhere (v6)
[17] 80 (v6) ALLOW IN Anywhere (v6)
```

Importance of Each Configured Rule:

1. Deny Incoming Traffic (Default): Blocks all unsolicited connections to keep the system secure from unauthorized access.
2. Allow Outgoing Traffic (Default): Ensures the system can communicate out, like downloading updates or accessing remote servers.
3. Allow SSH from Specific IP: Restricts remote management (SSH) to trusted IPs, reducing the risk of unauthorized access.
4. Deny SSH for Others: Blocks SSH access from all other sources to prevent attacks like brute force.
5. Allow HTTP/HTTPS: Opens ports 80 and 443 to enable secure web browsing or hosting, while keeping other services protected.

Testing: Validates the rules to confirm only the intended ports are accessible, ensuring the firewall works as expected.

Task 3

```
(root@kali) ~/home/kali
# sudo apt install apache2 mariadb-server php php-mysql php-gd libapache2-mod-php git -y

Package php-mysql is a virtual package provided by:
php8.2-mysql 8.2.21-1+b1
php7.4-mysql 7.4.3-4ubuntu2.24
You should explicitly select one to install.

Error: Package 'php-mysql' has no installation candidate

(root@kali)~/home/kali
# sudo service apache2 start
sudo service mysql start

(root@kali)~/home/kali
# sudo service apache2 status
sudo service mysql status

● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; disabled; preset: disabled)
   Active: active (running) since Sun 2024-12-08 15:25:42 EST; 10s ago
 Invocation: 148a5d7ad614e5c9591da88a181345c
    Docs: https://httpd.apache.org/docs/2.4/
   Process: 30724 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
  Main PID: 30748 (apache2)
    Tasks: 6 (limit: 3388)
   Memory: 32.1M (peak: 32.5M)
      CPU: 122ms
   CGroup: /system.slice/apache2.service
           └─30748 /usr/sbin/apache2 -k start
             └─30758 /usr/sbin/apache2 -k start
               └─30759 /usr/sbin/apache2 -k start
                 └─30760 /usr/sbin/apache2 -k start
                   └─30761 /usr/sbin/apache2 -k start
                     └─30762 /usr/sbin/apache2 -k start

Dec 08 15:25:41 kali systemd[1]: Starting apache2.service - The Apache HTTP Server...
Dec 08 15:25:42 kali apachectl[30739]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1.1. Set the 'ServerName' directive globally to suppress this message
Dec 08 15:25:42 kali systemd[1]: Started apache2.service - The Apache HTTP Server.
● mariadb.service - MariaDB 11.4.2 database server
   Loaded: loaded (/usr/lib/systemd/system/mariadb.service; disabled; preset: disabled)
   Active: active (running) since Sun 2024-12-08 15:00:23 EST; 25min ago
 Invocation: 92b34bed883f41fb636ea6284cf057
    Docs: man:mariadb(8)
          https://mariadb.com/kb/en/library/systemd/
   Process: 10082 ExecStartPre=/usr/bin/install -m 755 -o mysql -g root -d /var/run/mysqld (code=exited, status=0/SUCCESS)
   Process: 10084 ExecStartPre=/bin/sh -c systemctl unset-environment _WSREP_START_POSITION (code=exited, status=0/SUCCESS)

(root@kali)~/home/kali
# apt install apache2 mariadb-server php php-mysql libapache2-mod-php git

apache2 is already the newest version (2.4.62-1).
mariadb-server is already the newest version (1:11.4.2-4).
mariadb-server set to manually installed.
php is already the newest version (2:8.2+93+nmul).
php-mysql is already the newest version (2:8.2+93+nmul).
libapache2-mod-php is already the newest version (2:8.2+93+nmul).
libapache2-mod-php set to manually installed.
git is already the newest version (1:2.43.0-1+b1).
The following packages were automatically installed and are no longer required:
  dcraw flac libavfilter9 libbftw3-single3 libbsoncpp25 libbout123-0t64 libplacebo338 libpostproc57 libsynchron123-0t64 mpg123 rwho rwhod
Use 'sudo apt autoremove' to remove them.

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 1

(root@kali)~/home/kali
# cd /var/www/html; git clone https://github.com/digininja/DVWA.git

cd: too many arguments

(root@kali)~/home/kali
# cd /var/www/html
git clone https://github.com/digininja/DVWA.git

Cloning into 'DVWA' ...
remote: Enumerating objects: 4931, done.
remote: Counting objects: 100% (91/91), done.
remote: Compressing objects: 100% (35/35), done.
remote: Total 4931 (delta 52), reused 82 (delta 48), pack-reused 4840 (from 1)
Receiving objects: 100% (4931/4931), 2.45 MiB | 1.79 MiB/s, done.
Resolving deltas: 100% (2389/2389), done.

(root@kali)~/var/www/html
#
```

Click the mouse pointer inside or press Ctrl+G.

search

USD/RUB -0.45%


```
(root@kali) ~/home/kali
# sudo git clone https://github.com/digininja/DVWA.git /var/www/html/dvwa

Cloning into '/var/www/html/dvwa'...
remote: Enumerating objects: 4931, done.
remote: Counting objects: 100% (91/91), done.
remote: Compressing objects: 100% (35/35), done.
remote: Total 4931 (delta 52), reused 82 (delta 48), pack-reused 4840 (from 1)
Receiving objects: 100% (4931/4931), 2.41 MiB | 2.20 MiB/s, done.
Resolving deltas: 100% (2404/2404), done.

(root@kali) ~/home/kali
# sudo chmod -R www-data:www-data /var/www/html/dvwa
sudo chmod -R 755 /var/www/html/dvwa

(root@kali) ~/home/kali
# sudo nano /var/www/html/dvwa/config/config.inc.php

(root@kali) ~/home/kali
# sudo service apache2 restart

(root@kali) ~/home/kali
# sudo mysql -u root -p

Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 44
Server version: 11.4.2-MariaDB-4 Debian n/a

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Support MariaDB developers by giving a star at https://github.com/MariaDB/server
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

# the mouse pointer inside or press Ctrl+G
```

```
(root@kali) ~/home/kali
# sudo apt install apache2 mariadb-server php php-mysql php-gd libapache2-mod-php git -y

Package php-mysql is a virtual package provided by:
php8.2-mysql 8.2.21-1+b1
php7.4-mysql 7.4.3-4ubuntu2.24
You should explicitly select one to install.

Error: Package 'php-mysql' has no installation candidate

(root@kali) ~/home/kali
# sudo service apache2 start
sudo service mysql start

(root@kali) ~/home/kali
# sudo service apache2 status
sudo service mysql status

● apache2.service - The Apache HTTP Server
  Loaded: loaded (/usr/lib/systemd/system/apache2.service; disabled; preset: disabled)
  Active: active (running) since Sun 2024-12-08 15:25:42 EST; 10s ago
  Invocation: 148a5d7ad4614e5c9291da88c181345c
  Docs: https://httpd.apache.org/docs/2.4/
  Process: 30724 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
  Main PID: 30748 (apache2)
  Tasks: 6 (limit: 3388)
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  CPU: 122ms
  CGroup: /system.slice/apache2.service
          └─30748 /usr/sbin/apache2 -k start
            └─30758 /usr/sbin/apache2 -k start
              └─30759 /usr/sbin/apache2 -k start
                └─30760 /usr/sbin/apache2 -k start
                  └─30761 /usr/sbin/apache2 -k start
                    └─30762 /usr/sbin/apache2 -k start

Dec 08 15:25:41 kali systemd[1]: Starting apache2.service - The Apache HTTP Server...
Dec 08 15:25:42 kali apachectl[30739]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.0.1. Set the 'ServerName' directive globally to suppress this message
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● mariadb.service - MariaDB 11.4.2 database server
  Loaded: loaded (/usr/lib/systemd/system/mariadb.service; disabled; preset: disabled)
  Active: active (running) since Sun 2024-12-08 15:00:23 EST; 25min ago
  Invocation: 92b34bed883f4f1fb636ae284cfc057
  Docs: man:mariadb(8)
  https://mariadb.com/kb/en/library/systemd/
  Process: 10062 ExecStartPre=/usr/bin/install -m 755 -o mysql -g root -d /var/run/mysqld (code=exited, status=0/SUCCESS)
  Process: 10084 ExecStartPre=/bin/sh -c systemctl unset-environment _WSREP_START_POSITION (code=exited, status=0/SUCCESS)

(root@kali) ~/home/kali
# apt install apache2 mariadb-server php php-mysql libapache2-mod-php git

apache2 is already the newest version (2.4.62-1).
mariadb-server is already the newest version (1:11.4.2-4).
mariadb-server set to manually installed.
php is already the newest version (2:8.2+93+nmul).
php-mysql is already the newest version (2:8.2+93+nmul).
libapache2-mod-php is already the newest version (2:8.2+93+nmul).
libapache2-mod-php set to manually installed.
git is already the newest version (1:2.43.0-1+b1).
The following packages were automatically installed and are no longer required:
  dcraw flac libafl9 libffi2-single3 libjsoncpp25 libout123-0t64 libplacebo338 libpostproc57 libsyntax123-0t64 mpg123 rwho rwhod
Use 'sudo apt autoremove' to remove them.

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 1

(root@kali) ~/home/kali
# cd /var/www/html; git clone https://github.com/digininja/DVWA.git

cd: too many arguments

(root@kali) ~/home/kali
# cd /var/www/html
git clone https://github.com/digininja/DVWA.git

Cloning into 'DVWA'...
remote: Enumerating objects: 4931, done.
remote: Counting objects: 100% (91/91), done.
remote: Compressing objects: 100% (35/35), done.
remote: Total 4931 (delta 52), reused 82 (delta 48), pack-reused 4840 (from 1)
Receiving objects: 100% (4931/4931), 2.45 MiB | 1.79 MiB/s, done.
Resolving deltas: 100% (2389/2389), done.

(root@kali) ~/var/www/html
#

e the mouse pointer inside or press Ctrl+G.

b search

(root@kali) ~/home/kali
# sudo git clone https://github.com/digininja/DVWA.git /var/www/html/dvwa

Cloning into '/var/www/html/dvwa'...
remote: Enumerating objects: 4931, done.
remote: Counting objects: 100% (91/91), done.
remote: Compressing objects: 100% (35/35), done.
remote: Total 4931 (delta 52), reused 82 (delta 48), pack-reused 4840 (from 1)
Receiving objects: 100% (4931/4931), 2.41 MiB | 2.20 MiB/s, done.
Resolving deltas: 100% (2404/2404), done.

(root@kali) ~/home/kali
# sudo chown -R www-data:www-data /var/www/html/dvwa
sudo chmod -R 755 /var/www/html/dvwa

(root@kali) ~/home/kali
# sudo nano /var/www/html/dvwa/config/config.inc.php

(root@kali) ~/home/kali
# sudo service apache2 restart

(root@kali) ~/home/kali
# sudo mysql -u root -p

Enter password:
Welcome to the MariaDB monitor. Commands end with ; or \g.
Your MariaDB connection id is 44
Server version: 11.4.2-MariaDB-4 Debian n/a

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Support MariaDB developers by giving a star at https://github.com/MariaDB/server
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.
```

```

root@kali: ~# /etc/snort
root@kali: ~# snort -A console -g -u snort -g snort -c /etc/snort/snort.conf -i eth1
09/27-03:50:01.285015 ** [1:1000000:0] "ICMP message found" ** [Priority: 0] {ICMP} 150.1.7.101 → 150.1.7.104
09/27-03:50:01.285015 ** [1:1000007:1] Ping of Death detected ** [Priority: 0] {ICMP} 150.1.7.101 → 150.1.7.104
09/27-03:50:01.285015 ** [1:1000004:1] Smurf attack detected ** [Priority: 0] {ICMP} 150.1.7.101 → 150.1.7.104
09/27-03:50:02.289055 ** [1:1000000:0] "ICMP message found" ** [Priority: 0] {ICMP} 150.1.7.101 → 150.1.7.104
09/27-03:50:02.289055 ** [1:1000007:1] Ping of Death detected ** [Priority: 0] {ICMP} 150.1.7.101 → 150.1.7.104
09/27-03:50:02.289055 ** [1:1000004:1] Smurf attack detected ** [Priority: 0] {ICMP} 150.1.7.101 → 150.1.7.104
09/27-03:50:02.309160 ** [1:1000000:0] "ICMP message found" ** [Priority: 0] {ICMP} 150.1.7.104 → 150.1.7.101
09/27-03:50:03.290870 ** [1:1000000:0] "ICMP message found" ** [Priority: 0] {ICMP} 150.1.7.101 → 150.1.7.104
09/27-03:50:03.290870 ** [1:1000007:1] Ping of Death detected ** [Priority: 0] {ICMP} 150.1.7.101 → 150.1.7.104
09/27-03:50:03.290870 ** [1:1000004:1] Smurf attack detected ** [Priority: 0] {ICMP} 150.1.7.101 → 150.1.7.104
09/27-03:50:03.292866 ** [1:1000000:0] "ICMP message found" ** [Priority: 0] {ICMP} 150.1.7.104 → 150.1.7.101
09/27-03:50:04.293599 ** [1:1000000:0] "ICMP message found" ** [Priority: 0] {ICMP} 150.1.7.101 → 150.1.7.104
09/27-03:50:04.293599 ** [1:1000007:1] Ping of Death detected ** [Priority: 0] {ICMP} 150.1.7.101 → 150.1.7.104
09/27-03:50:04.293599 ** [1:1000004:1] Smurf attack detected ** [Priority: 0] {ICMP} 150.1.7.101 → 150.1.7.104
09/27-03:50:04.296303 ** [1:1000000:0] "ICMP message found" ** [Priority: 0] {ICMP} 150.1.7.104 → 150.1.7.101
09/27-03:50:05.296053 ** [1:1000000:0] "ICMP message found" ** [Priority: 0] {ICMP} 150.1.7.101 → 150.1.7.104
09/27-03:50:05.296053 ** [1:1000007:1] Ping of Death detected ** [Priority: 0] {ICMP} 150.1.7.101 → 150.1.7.104
09/27-03:50:05.296053 ** [1:1000004:1] Smurf attack detected ** [Priority: 0] {ICMP} 150.1.7.101 → 150.1.7.104
09/27-03:50:05.298227 ** [1:1000000:0] "ICMP message found" ** [Priority: 0] {ICMP} 150.1.7.104 → 150.1.7.101
09/27-03:50:06.297910 ** [1:1000000:0] "ICMP message found" ** [Priority: 0] {ICMP} 150.1.7.101 → 150.1.7.104
09/27-03:50:06.297910 ** [1:1000007:1] Ping of Death detected ** [Priority: 0] {ICMP} 150.1.7.101 → 150.1.7.104
09/27-03:50:06.297910 ** [1:1000004:1] Smurf attack detected ** [Priority: 0] {ICMP} 150.1.7.101 → 150.1.7.104
09/27-03:50:06.299856 ** [1:1000000:0] "ICMP message found" ** [Priority: 0] {ICMP} 150.1.7.104 → 150.1.7.101
09/27-03:50:07.297838 ** [1:1000000:0] "ICMP message found" ** [Priority: 0] {ICMP} 150.1.7.101 → 150.1.7.104
09/27-03:50:07.297838 ** [1:1000007:1] Ping of Death detected ** [Priority: 0] {ICMP} 150.1.7.101 → 150.1.7.104
09/27-03:50:07.297838 ** [1:1000004:1] Smurf attack detected ** [Priority: 0] {ICMP} 150.1.7.101 → 150.1.7.104
09/27-03:50:07.299893 ** [1:1000000:0] "ICMP message found" ** [Priority: 0] {ICMP} 150.1.7.104 → 150.1.7.101
09/27-03:50:08.300124 ** [1:1000000:0] "ICMP message found" ** [Priority: 0] {ICMP} 150.1.7.101 → 150.1.7.104
09/27-03:50:08.300124 ** [1:1000007:1] Ping of Death detected ** [Priority: 0] {ICMP} 150.1.7.101 → 150.1.7.104

```

```

GNU nano 8.1 /etc/snort/rules/local.rules
# $Id: local.rules,v 1.11 2004/07/23 20:15:44 bmc Exp $
#
# LOCAL RULES
#
# This file intentionally does not come with signatures. Put your local
# additions here.
alert icmp any any → 150.1.7.0/24 any (msg: "ICMP message found";sid:1000000;)
alert tcp any any → 150.1.7.0/24 80 (msg:"Potential SYN Flood Attack"; flags:S; threshold:type threshold, track by_src, count
alert tcp 150.1.7.0/24 any → 150.1.7.0/24 any (msg:"LAND Attack Detected"; flags:S; dsize:0; sid:1000003; rev:1;)
alert icmp 150.1.7.0/24 any → 150.1.7.0/24 any (msg:"Smurf attack detected"; itype:8; sid:1000004; rev:1;)
alert udp 150.1.7.0/24 any → 150.1.7.0/24 53 (msg:"DNS Poisoning Attack Detected"; content:"00 01 00 01"; offset:2; depth
alert udp any any → 150.1.7.0/24 53 (msg:"DNS Amplification Attack Detected"; threshold:type limit, track by_src, count 10,
alert icmp any any → 150.1.7.0/24 any (msg:"Ping of Death detected"; itype:8; dsize:>1400; sid:1000007; rev:1;)

```

```

root@kali: ~# /etc/snort
root@kali: ~# snort -A console -g -u snort -g snort -c /etc/snort/snort.conf -i eth1
09/27-03:31:49.631923 ** [1:1000006:1] DNS Amplification Attack Detected ** [Priority: 0] {UDP} 150.1.7.101:2657 → 150.1.7.104:53
09/27-03:31:50.632861 ** [1:1000006:1] DNS Amplification Attack Detected ** [Priority: 0] {UDP} 150.1.7.101:2658 → 150.1.7.104:53
09/27-03:31:51.634115 ** [1:1000006:1] DNS Amplification Attack Detected ** [Priority: 0] {UDP} 150.1.7.101:2659 → 150.1.7.104:53
09/27-03:31:52.635324 ** [1:1000006:1] DNS Amplification Attack Detected ** [Priority: 0] {UDP} 150.1.7.101:2660 → 150.1.7.104:53
09/27-03:31:53.636800 ** [1:1000006:1] DNS Amplification Attack Detected ** [Priority: 0] {UDP} 150.1.7.101:2661 → 150.1.7.104:53
09/27-03:31:54.636827 ** [1:1000006:1] DNS Amplification Attack Detected ** [Priority: 0] {UDP} 150.1.7.101:2662 → 150.1.7.104:53
09/27-03:31:55.637702 ** [1:1000006:1] DNS Amplification Attack Detected ** [Priority: 0] {UDP} 150.1.7.101:2663 → 150.1.7.104:53
09/27-03:31:56.638423 ** [1:1000006:1] DNS Amplification Attack Detected ** [Priority: 0] {UDP} 150.1.7.101:2664 → 150.1.7.104:53
09/27-03:31:57.639567 ** [1:1000006:1] DNS Amplification Attack Detected ** [Priority: 0] {UDP} 150.1.7.101:2665 → 150.1.7.104:53
09/27-03:31:58.640667 ** [1:1000006:1] DNS Amplification Attack Detected ** [Priority: 0] {UDP} 150.1.7.101:2666 → 150.1.7.104:53
09/27-03:31:59.641891 ** [1:1000006:1] DNS Amplification Attack Detected ** [Priority: 0] {UDP} 150.1.7.101:2667 → 150.1.7.104:53
09/27-03:32:00.642317 ** [1:1000006:1] DNS Amplification Attack Detected ** [Priority: 0] {UDP} 150.1.7.101:2668 → 150.1.7.104:53
09/27-03:32:01.643136 ** [1:1000006:1] DNS Amplification Attack Detected ** [Priority: 0] {UDP} 150.1.7.101:2669 → 150.1.7.104:53
09/27-03:32:02.643670 ** [1:1000006:1] DNS Amplification Attack Detected ** [Priority: 0] {UDP} 150.1.7.101:2670 → 150.1.7.104:53
09/27-03:32:03.644750 ** [1:1000006:1] DNS Amplification Attack Detected ** [Priority: 0] {UDP} 150.1.7.101:2671 → 150.1.7.104:53
09/27-03:32:04.645475 ** [1:1000006:1] DNS Amplification Attack Detected ** [Priority: 0] {UDP} 150.1.7.101:2672 → 150.1.7.104:53
09/27-03:32:05.646590 ** [1:1000006:1] DNS Amplification Attack Detected ** [Priority: 0] {UDP} 150.1.7.101:2673 → 150.1.7.104:53
09/27-03:32:06.647507 ** [1:1000006:1] DNS Amplification Attack Detected ** [Priority: 0] {UDP} 150.1.7.101:2674 → 150.1.7.104:53
09/27-03:32:07.648407 ** [1:1000006:1] DNS Amplification Attack Detected ** [Priority: 0] {UDP} 150.1.7.101:2675 → 150.1.7.104:53
09/27-03:32:08.649145 ** [1:1000006:1] DNS Amplification Attack Detected ** [Priority: 0] {UDP} 150.1.7.101:2676 → 150.1.7.104:53
09/27-03:32:09.650072 ** [1:1000006:1] DNS Amplification Attack Detected ** [Priority: 0] {UDP} 150.1.7.101:2677 → 150.1.7.104:53
09/27-03:32:10.651200 ** [1:1000006:1] DNS Amplification Attack Detected ** [Priority: 0] {UDP} 150.1.7.101:2678 → 150.1.7.104:53
09/27-03:32:11.651982 ** [1:1000006:1] DNS Amplification Attack Detected ** [Priority: 0] {UDP} 150.1.7.101:2679 → 150.1.7.104:53
09/27-03:32:12.652822 ** [1:1000006:1] DNS Amplification Attack Detected ** [Priority: 0] {UDP} 150.1.7.101:2680 → 150.1.7.104:53
09/27-03:32:13.653657 ** [1:1000006:1] DNS Amplification Attack Detected ** [Priority: 0] {UDP} 150.1.7.101:2681 → 150.1.7.104:53
09/27-03:32:14.654804 ** [1:1000006:1] DNS Amplification Attack Detected ** [Priority: 0] {UDP} 150.1.7.101:2682 → 150.1.7.104:53
09/27-03:32:15.655763 ** [1:1000006:1] DNS Amplification Attack Detected ** [Priority: 0] {UDP} 150.1.7.101:2683 → 150.1.7.104:53
09/27-03:32:16.656558 ** [1:1000006:1] DNS Amplification Attack Detected ** [Priority: 0] {UDP} 150.1.7.101:2684 → 150.1.7.104:53
09/27-03:32:17.657696 ** [1:1000006:1] DNS Amplification Attack Detected ** [Priority: 0] {UDP} 150.1.7.101:2685 → 150.1.7.104:53
09/27-03:32:18.658777 ** [1:1000006:1] DNS Amplification Attack Detected ** [Priority: 0] {UDP} 150.1.7.101:2686 → 150.1.7.104:53
09/27-03:32:19.659493 ** [1:1000006:1] DNS Amplification Attack Detected ** [Priority: 0] {UDP} 150.1.7.101:2687 → 150.1.7.104:53
09/27-03:32:20.660551 ** [1:1000006:1] DNS Amplification Attack Detected ** [Priority: 0] {UDP} 150.1.7.101:2688 → 150.1.7.104:53

```



```

# LOCAL RULES
#
# This file intentionally does not come with signatures. Put your local
# additions here.
alert icmp any any → 150.1.7.0/24 any (msg: "ICMP message found";sid:10000001;)
alert tcp any any → 150.1.7.0/24 80 (msg:"Potential SYN Flood Attack"; flags:S; threshold:type threshold, track by_src, count 100, seconds 10)
alert tcp 150.1.7.0/24 any → 150.1.7.0/24 any (msg:"LAND Attack Detected"; flags:S; dsize:0; sid:10000003; rev:1;)
alert icmp 150.1.7.0/24 any → 150.1.7.0/24 any (msg:"Smurf attack detected"; itype:8; sid:10000004; rev:1;)
alert udp 150.1.7.0/24 any → 150.1.7.0/24 53 (msg:"DNS Poisoning Attack Detected"; content:"\00 01 00 01"; offset:2; depth:4; sid:10000005;
alert udp any any → 150.1.7.0/24 53 (msg:"DNS Amplification Attack Detected"; threshold:type limit, track by_src, count 10, seconds 1; sid:10

# snort -A console -q -u snort -c /etc/snort/snort.conf -i eth1
09/27-03:03:32.893826 ** [1:10000001:0] "ICMP message found" ** [Priority: 0] [ICMP] 150.1.7.104 → 150.1.7.255
09/27-03:03:32.893826 ** [1:10000004:1] Smurf attack detected ** [Priority: 0] [ICMP] 150.1.7.104 → 150.1.7.255
09/27-03:03:33.893971 ** [1:10000001:0] "ICMP message found" ** [Priority: 0] [ICMP] 150.1.7.104 → 150.1.7.255
09/27-03:03:33.893971 ** [1:10000004:1] Smurf attack detected ** [Priority: 0] [ICMP] 150.1.7.104 → 150.1.7.255
09/27-03:03:34.895171 ** [1:10000001:0] "ICMP message found" ** [Priority: 0] [ICMP] 150.1.7.104 → 150.1.7.255
09/27-03:03:34.895171 ** [1:10000004:1] Smurf attack detected ** [Priority: 0] [ICMP] 150.1.7.104 → 150.1.7.255
09/27-03:03:35.896500 ** [1:10000001:0] "ICMP message found" ** [Priority: 0] [ICMP] 150.1.7.104 → 150.1.7.255
09/27-03:03:35.896500 ** [1:10000004:1] Smurf attack detected ** [Priority: 0] [ICMP] 150.1.7.104 → 150.1.7.255
09/27-03:03:36.897300 ** [1:10000001:0] "ICMP message found" ** [Priority: 0] [ICMP] 150.1.7.104 → 150.1.7.255
09/27-03:03:36.897300 ** [1:10000004:1] Smurf attack detected ** [Priority: 0] [ICMP] 150.1.7.104 → 150.1.7.255
09/27-03:03:37.899015 ** [1:10000001:0] "ICMP message found" ** [Priority: 0] [ICMP] 150.1.7.104 → 150.1.7.255
09/27-03:03:37.899015 ** [1:10000004:1] Smurf attack detected ** [Priority: 0] [ICMP] 150.1.7.104 → 150.1.7.255
09/27-03:03:38.915961 ** [1:10000001:0] "ICMP message found" ** [Priority: 0] [ICMP] 150.1.7.104 → 150.1.7.255
09/27-03:03:38.915961 ** [1:10000004:1] Smurf attack detected ** [Priority: 0] [ICMP] 150.1.7.104 → 150.1.7.255
09/27-03:03:39.917681 ** [1:10000001:0] "ICMP message found" ** [Priority: 0] [ICMP] 150.1.7.104 → 150.1.7.255
09/27-03:03:39.917681 ** [1:10000004:1] Smurf attack detected ** [Priority: 0] [ICMP] 150.1.7.104 → 150.1.7.255
09/27-03:03:40.919196 ** [1:10000001:0] "ICMP message found" ** [Priority: 0] [ICMP] 150.1.7.104 → 150.1.7.255
09/27-03:03:40.919196 ** [1:10000004:1] Smurf attack detected ** [Priority: 0] [ICMP] 150.1.7.104 → 150.1.7.255
09/27-03:03:41.920492 ** [1:10000001:0] "ICMP message found" ** [Priority: 0] [ICMP] 150.1.7.104 → 150.1.7.255
09/27-03:03:41.920492 ** [1:10000004:1] Smurf attack detected ** [Priority: 0] [ICMP] 150.1.7.104 → 150.1.7.255
09/27-03:03:42.921391 ** [1:10000001:0] "ICMP message found" ** [Priority: 0] [ICMP] 150.1.7.104 → 150.1.7.255
09/27-03:03:42.921391 ** [1:10000004:1] Smurf attack detected ** [Priority: 0] [ICMP] 150.1.7.104 → 150.1.7.255
09/27-03:03:43.922164 ** [1:10000001:0] "ICMP message found" ** [Priority: 0] [ICMP] 150.1.7.104 → 150.1.7.255
09/27-03:03:43.922164 ** [1:10000004:1] Smurf attack detected ** [Priority: 0] [ICMP] 150.1.7.104 → 150.1.7.255
09/27-03:03:44.924266 ** [1:10000001:0] "ICMP message found" ** [Priority: 0] [ICMP] 150.1.7.104 → 150.1.7.255
09/27-03:03:44.924266 ** [1:10000004:1] Smurf attack detected ** [Priority: 0] [ICMP] 150.1.7.104 → 150.1.7.255
09/27-03:03:45.925697 ** [1:10000001:0] "ICMP message found" ** [Priority: 0] [ICMP] 150.1.7.104 → 150.1.7.255
09/27-03:03:45.925697 ** [1:10000004:1] Smurf attack detected ** [Priority: 0] [ICMP] 150.1.7.104 → 150.1.7.255
09/27-03:03:46.927041 ** [1:10000001:0] "ICMP message found" ** [Priority: 0] [ICMP] 150.1.7.104 → 150.1.7.255
09/27-03:03:46.927041 ** [1:10000004:1] Smurf attack detected ** [Priority: 0] [ICMP] 150.1.7.104 → 150.1.7.255
09/27-03:03:47.928213 ** [1:10000001:0] "ICMP message found" ** [Priority: 0] [ICMP] 150.1.7.104 → 150.1.7.255
09/27-03:03:47.928213 ** [1:10000004:1] Smurf attack detected ** [Priority: 0] [ICMP] 150.1.7.104 → 150.1.7.255

```

```
File Actions Edit View Help
GNU nano 8.1 jadu.py
from scapy.all import *

# Define the target IP
target_ip = "150.1.7.104"

# Create an ICMP packet
icmp_packet = IP(dst=target_ip)/ICMP()

# Send the packet and wait for a response
response = sr1(icmp_packet, timeout=2)

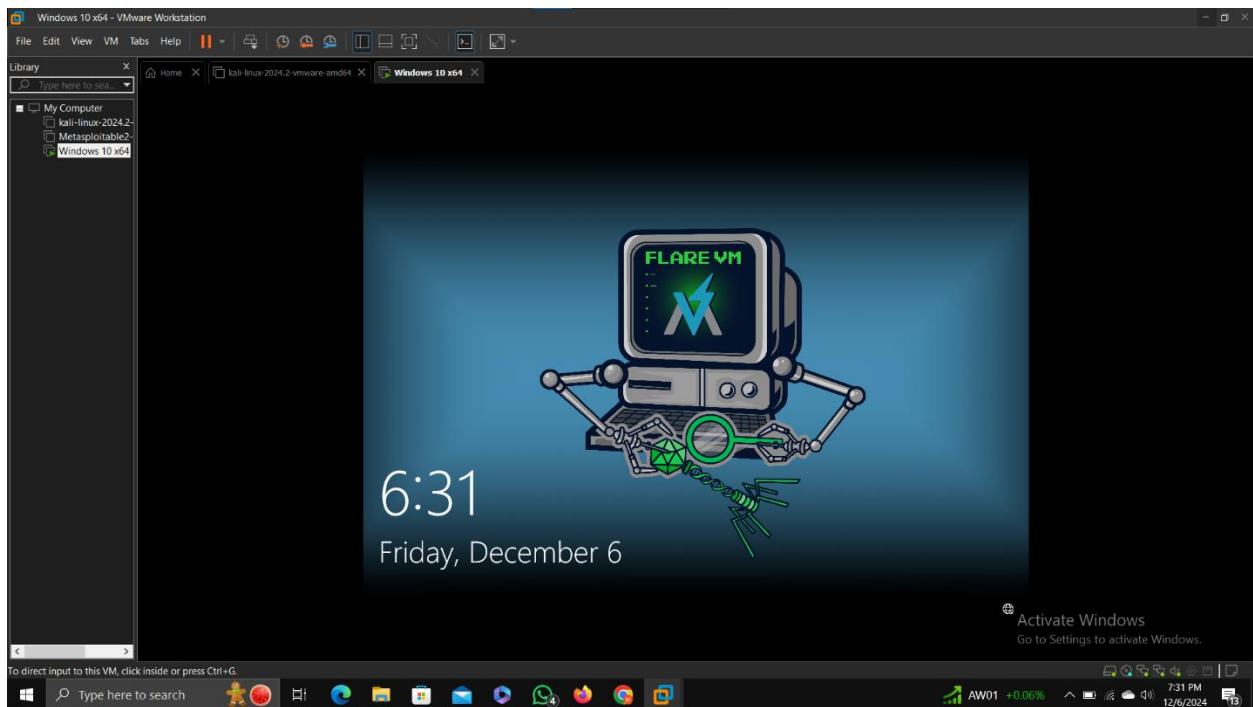
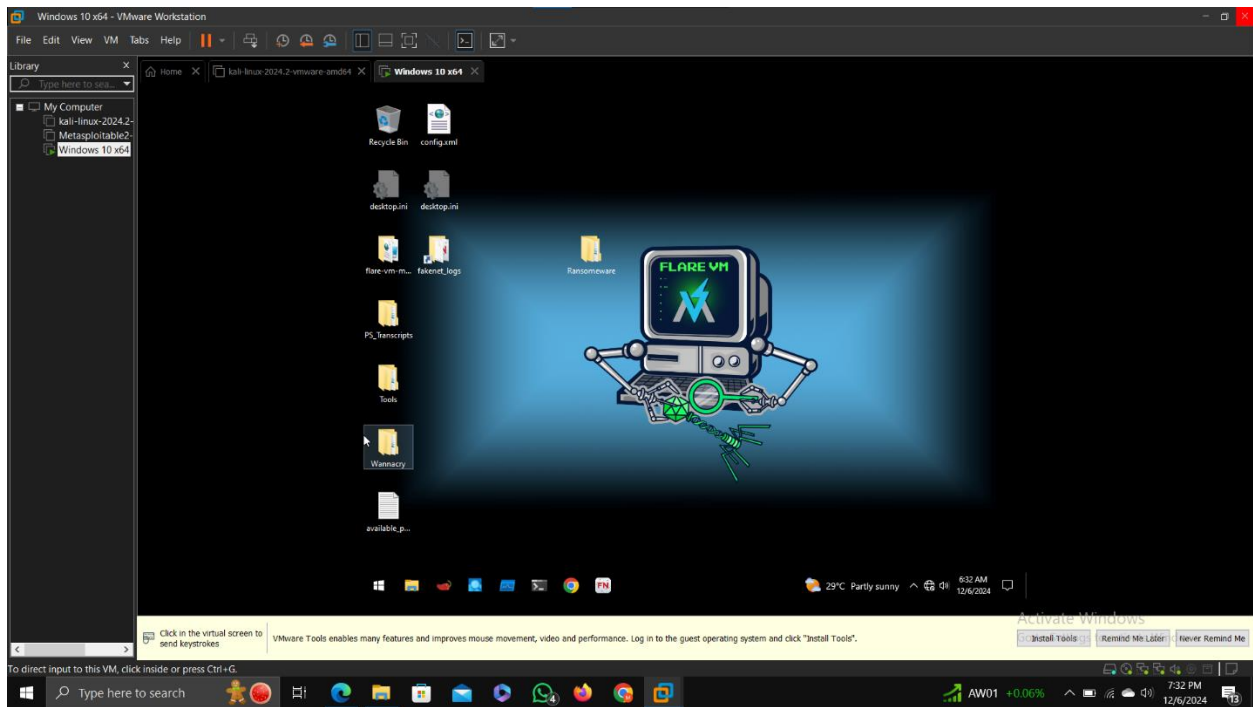
# Check if we received a response
if response:
    print(f"Received response from {target_ip}:")
    response.show()
else:
    print(f"No response from {target_ip}.")

[ Read 17 lines ]
at LOIC.frnMain.Dispose (System.Boolean disposing) [0x0001f] in <26bc7b3a44044bd78642121ecc4f60>
at System.ComponentModel.Component.Finalize () [0x00000] in <a85c1a570f9a4f9f9c3d2cfa5504e34f>:
(root@kali)~/home/kali/loicinstaller/LOIC
$ ./loic.sh run
/usr/bin/mono
cp: warning: behavior of -n is non-portable and may change in future; use --update=none instead
^C
```



```
Forms.Form.OnLayout (System.Windows.Forms.LayoutEventArgs levent) [0x000b0] in
>:0
F
F
F
0
n
F
2
F
6
F
F
S
t
me/kali/loicinstaller/LOIC
N O N E I
# hping3 --flood --udp -p 53 150.1.7.104
HPING 150.1.7.104 (eth1 150.1.7.104): udp mode set, 28 headers + 0 data bytes
hping in flood mode, no replies will be shown
```

Task4



```
PS C:\Windows\system32> Set-ExecutionPolicy Unrestricted
```

```
PS C:\Users\PC\Desktop> cd flare-vm-main  
PS C:\Users\PC\Desktop\flare-vm-main> █
```

Types:

Static Analysis: examining the malware without executing it.

Techniques include:

Disassembly: Breaking down the executable code into assembly instruction

Decompilation: converting assembly code into higher level programming language

String Analysis.: searching for textual strings (url, links) within malware.

Dynamic Analysis: involves running the malware in a controlled environment (sandbox) to observe its behaviour.

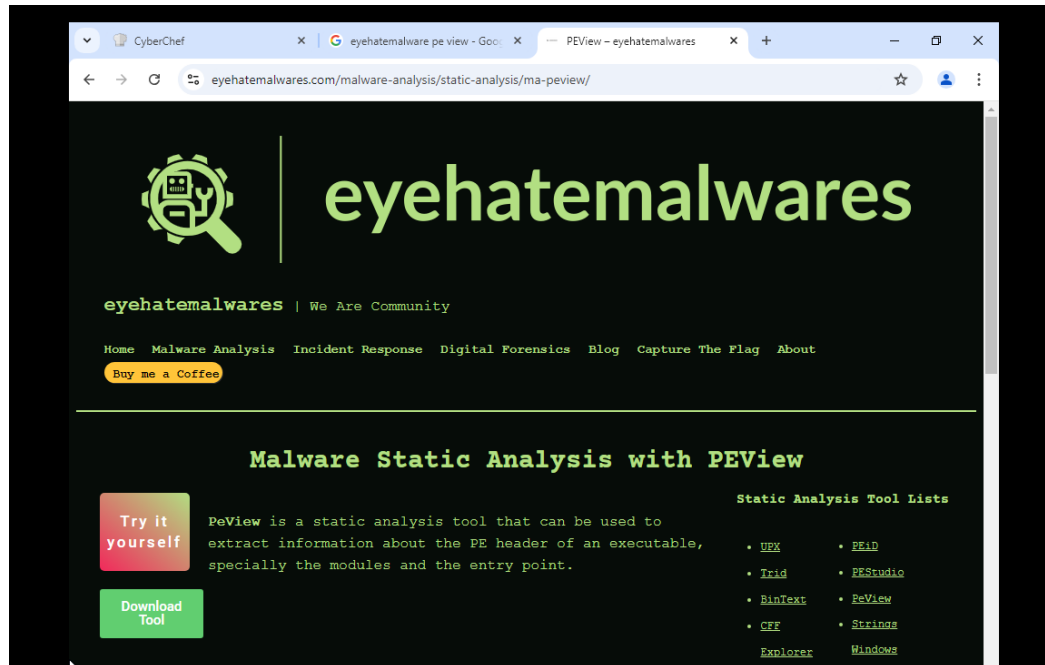
Technique:

debugging: step by step execution of the malware to understand its action.

Behavioural Analysis: monitoring the malware interaction between

system and network

Memory analysis: memory footprint of malware.



Task 5

example of an insecure Python script:

```
import os
```

```
def connect_to_database():
```

```
    # Hardcoded credentials (vulnerability)
```

```
    username = "admin"
```

```
    password = "password123"
```

```
    connection_string=f"mysql://{username}:{password}@localhost/mydatabase"
```

```
    return connection_string
```

```
def execute_command(command):
```

```
    # Unsantized input (vulnerability)
```

```
    os.system(command)
```

Review the Code for Vulnerabilities

1. **Hardcoded Credentials:** Storing credentials directly in the code is insecure because it exposes sensitive information.
2. **Unsantized Inputs:** Using `os.system` with unsantized input allows for command injection attacks.

Fix the Code

Fixing Hardcoded Credentials

- **Solution:** Use environment variables to store credentials securely.

Fixed Python Code:

```
import os
```

```
def connect_to_database():
```

```
    # Retrieve credentials from environment variables
```

```
    username = os.getenv('DB_USERNAME')
```

```
    password = os.getenv('DB_PASSWORD')
```

```
    connection_string=f'mysql://{username}:{password}@localhost/mydatabase'
```

```
    return connection_string
```

Fixing Unsantized Inputs

Fixed Python Code:

```
import os
```

```
def execute_command(command):
```

```
    # Sanitize input (example: only allow specific commands)
```

```
    allowed_commands = ["ls", "whoami"]
```

```
    if command in allowed_commands:
```

```
        os.system(command)
```

```
    else:
```

```
        print("Invalid command")
```

Insecure Practices:

1. **Hardcoded Credentials:** Storing credentials directly in the source code can expose sensitive information if the code is ever accessed by unauthorized users.
2. **Unsanitized Inputs:** Allowing user inputs to be executed directly by the system without validation can lead to command injection attacks.

How They Were Resolved:

1. **Hardcoded Credentials:** Replaced hardcoded credentials with environment variables. This ensures that sensitive information is not exposed in the source code.
2. **Unsanitized Inputs:** Added input validation and sanitized the commands. Only allowed commands from a predefined list are executed, preventing command injection.